INCH-POUND
A-A-60021
26 May 2021

COMMERCIAL ITEM DESCRIPTION

COMPUTED (DIGITAL) RADIOGRAPHY IMAGING SYSTEM

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE This commercial item description provides the minimum requirements for a computed radiography (CR) digital radiographic imaging system suitable for nondestructive inspection of a variety of metallic and non-metallic products used in military aircraft maintenance environments, which may be located at fixed military bases or deployed to austere forward operating locations. This document contains system specifications necessary to meet Department of Defense (DoD) requirements for radiographic inspections such as foreign object detection, water entrapment, fatigue crack detection, and welder certification.

2. CLASSIFICATION. The system described in this document is classified as a complete CR system, employing state-of-the-art storage phosphor imaging plates (IP) as the collection media for radiographic data used for nondestructive inspection/testing/evaluation. An automated, programmable method of extracting the data and subsequently erasing the IP, and a method for displaying and storing the images shall be provided.

2.1 CR system. The CR system shall be a table-top, portable system. The CR system will contain an integral eraser capable of processing industry standard flexible IPs, from 5 x 7 inch to 14 x 17 inch, as well as custom shape flexible IPs as small as 2 x 4 inch. Metric equivalent IP sizes are acceptable.

3 SALIENT CHARACTERISTICS.

3.1 System. The CR system described in this document shall be a turnkey system and shall include the following:

a. Flexible phosphor IPs.

b. Flexible exposure cassettes.

c. Rigid cassettes or adapters for processing, if required.

d. Scanner (Image Reader/Erasure Unit/Processing Cassettes).

e. System Controller/Workstation.

f. Display Monitor(s) for image acquisition, image viewing, database management, data

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data that may improve this document should be sent to: AFLCMC/EZSS, 2145 Monahan Way, Area B, Bldg 28, Wright-Patterson AFB, OH 45433 or email to ENGINEERING.STANDARDS@US.AF.MIL. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:32Z
Check the source to verify that this is the current version before use.

archiving, data sharing, and printing.

g. Hard-shell, appropriately padded, shipping containers/carrying cases suitable for storing and shipping the unit with accessories for all system components and accessories.

h. Process control devices.

3.2. Deliverables. Table I contains a list of deliverables for this document.

3.3. System capabilities. The CR System shall be capable of producing radiographic images equal to or exceeding the general radiographic requirements of the American Standards for Testing and Materials (ASTM) specification ASTM-E1742/E1742M, Standard Practice for Radiographic Examination and E2033, Standard Practice for Radiographic Examination Using Computed Radiography (Photostimulable Luminescence Method).

3.3.1 Environment. The CR system shall be capable of operating in an industrial environment. Suitable mechanisms shall be incorporated to protect hardware, including IPs, from damage due to airborne particulates. All proposals shall describe these protective measures and testing by an independent laboratory to MIL-STD-810H, paragraph 510.7, low velocity sand and dust, or equivalent. Prior testing accomplished to a previous revision of MIL- STD-810 is acceptable, but shall be identified in the proposal.

3.3.1.1 System operating temperature requirements. The CR system shall be capable of performing to the requirements of this CID throughout temperatures from 50 to 110 degrees Fahrenheit at 90 percent relative humidity, and from 0 to 7,000 feet above sea level. All proposals shall include testing to MIL-PRF-28800F, paragraph 3.8.2, or equivalent, testing by an independent laboratory. Prior testing accomplished to a previous revision of MIL-PRF-28800 is acceptable, but shall be identified in the proposal.

3.3.1.2. Storage temperature requirements. The CR system shall be capable of being stored at temperatures from 0 to 150 degrees Fahrenheit at 90 percent relative humidity. All proposals shall include testing to MIL-PRF-28800F, paragraph 3.8.2.1, or equivalent testing by an independent laboratory. Prior testing accomplished to a previous revision of MIL-PRF-28800 is acceptable, but shall be identified in the proposal.

3.3.1.3 Power requirements. The CR system shall be capable of operation when supplied with 100-250 volt alternating current - 50 or 60 Hertz (100-250 VAC - 50/60 Hz) automatically selected.

3.3.2 Performance. The CR system shall be capable of scanning and erasing a minimum of 15, 14 x 17 inch IPs per hour at 50 micron sampling resolution.

3.3.3 Size and weight. The complete CR system in shipping cases shall not exceed a volume of 125 cubic feet, with no dimension larger than five (5) feet. The complete CR system in shipping cases shall weigh no more than 500 pounds.

3.3.4 Imaging plates and cassettes.

3.3.4.1 IP bend radius. All flexible IPs shall have a minimum bend radius of three inches without damage.

3.3.4.2 General purpose IPs. The following general purpose IPs shall be included: two, 14 x 17 inch flexible IPs, two, 8 x 10 inch flexible IPs, and two, 5 x 7 inch flexible IPs. For this document, a general purpose IP shall be able to visually resolve 2.5 line pairs per millimeter on both gauges on the process control device described in 3.8.2.

3.3.4.3 High resolution IPs. The following high resolution IPs shall be included: five, 14 x 17

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:32Z
Check the source to verify that this is the current version before use.

inch flexible IPs; two, 8 x 10 inch flexible IPs; and two, 5 x 7 inch flexible IPs. For this document, a high resolution IP shall be able to visually resolve 6.0 line pairs per millimeter on both gauges on the process control device described in 3.8.2.

3.3.4.4 Ultra-high resolution IPs. The following ultra-high resolution IPs (Note these are often referred to as "blue" IPs.) shall be included: two, 8 x 10 inch flexible IPs, and two, 5 x 7 inch flexible IPs. For this document, an ultra-high resolution IP shall be able to visually resolve 8.0 line pairs per millimeter on both gauges on the process control device described in 3.8.2.

3.3.4.5 Flexible cassettes. Seven, 14 x 17 inch; six, 8 x 10 inch; and six, 5 x 7 inch flexible cassettes shall be included.

3.3.5 IP cleaning system. Manufacturer recommended imaging plate cleaning system to include fluid, wiping cloths, etc., necessary to clean 100, 14x17 inch IPs shall be included.

3.3.6 Maintenance kit. Manufacturer one-time recommended maintenance kit to include all hardware, supplies, tools, etc., shall be included.

3.3.7 Scanner (Image reader/erasure unit). Drum scanners are specifically excluded from consideration due to USAF history of operational damage to imaging plates. Drum scanners require the imaging plate to be placed active-side down on a half-cylinder and then mechanically moved over the laser-scanning window. Any particulates (e.g., dust, sand, etc.,) on the drum cylinder or imaging plate will cause scratches to the imaging plate, causing an unacceptable decrease in the operational life of the imaging plate and increase in life-cycle costs. The scanner shall:

a Operate with commercially available IPs noted in 3.3.4, including prior-generation IPs.

b. Be capable of automatically reading and erasing the IP in a single operation.

c. Read the IP at a 14-Bit minimum gray scale depth and use a linear lookup table (LUT) to display the data.

d. Feature integral diagnostics and communication of error conditions with the workstation.

e. Be capable of acquiring data from the IP at 50 and 100 micron sampling resolution at a minimum.

f. Be capable of handling "cut" IPs (i.e. IPs of non-standard, unique shapes). It is acceptable to use a special or custom adapter to scan "cut" IPs and a minimum of two adapters shall be included.

g. Include processing cassettes required for directly loading the IP in the scanner. If the processing cassette is not integral to the scanner, a minimum of two processing cassettes and required accessories shall be provided to accommodate the IP sizes listed in 3.3.4. Any special tools required to insert/extract an IP in the processing cassette shall be included.

h. Include an erase function that may be operated independent from the scan/erase cycle.

3.3.8 System controller/workstation. The workstation shall include a CPU, corded keyboard and mouse, high resolution monochrome display monitor, and secondary monitor if necessary.

3.3.8.1 Workstation hardware. The workstation hardware shall be capable of supporting the overall CR system functionality and technical requirements (i.e. I.O, processor speed, video graphics, memory, etc.).

3.3.8.2. Video output. The workstation shall also contain a minimum of one of each of the following: a separate connection for a VGA projector for training, HMDI, or other high definition

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:32Z
Check the source to verify that this is the current version before use.

output port and color display monitor for file maintenance; this projector or monitor is not a contract deliverable.

3.3.8.3 Hardware key. Hardware type software keys or dongles shall not be required for workstation/software functionality or license verification.

3.3.8.4 Workstation control. The workstation shall serve as the controlling programming device and driver system for the scanner (image reader/eraser unit) and shall serve as a viewing device to view, manipulate, interpret, and store images.

3.3.8.5 Workstation accessories. The workstation shall include:

a. An internal dual-format CD-DVD R/W where content can be viewed on other DVD-ROM drives, DVD players and DVD recorders.

b. Sufficient USB ports available for transferring data to/from external devices after all peripheral devices are connected.

c. Two internal hard drives in a Redundant Array of Independent Disks (RAID 1) (mirrored) configuration, each with a minimum of two terabyte (2 TB) storage capacity. Separate hard drive partitions for system files, system recovery, and image storage may be used.

d. Cables necessary to connect to the scanner and monitor.

3.3.9 Display monitor. The CR system shall contain one monochrome, display monitor with a minimum 21 inch diagonal measurement, flat panel, five mega pixel (minimum resolution) for viewing the computed radiograph. Minimum resolution shall be 2560 by 2048 pixels. Additional monitors shall be included if necessary to operate the system.

3.3.10 Software, operating system. The system controller/workstation shall use a Windows 10 (64-bit) Operating System (OS).

3.3.11 Software, application. All delivered imaging and scanner control software shall be of the same version/build during the duration of the contract. Delivered imaging and scanner control software shall be the latest approved commercial version at contract award.

3.3.11.1 Software updates. All software updates made by the manufacturer during the duration of the contract shall be included at no additional charge.

3.3.11.2 Software-configuration control. All proposals shall include detailed recommended procedures for maintaining configuration control of software.

3.3.11.3 Software update. All software updates shall be coordinated with the Cognizant Engineer. The OEM shall not issue or distribute software updates directly to field users without this coordination.

3.3.12 Print functionality. The CR system shall have the capability to print from typical office laser or ink-jet printers, as well as other specialty DICONDE or Digital Imaging and Communications in Medicine (DICOM) compatible printers offered in commercial CR systems.

3.3.13 Software deliverables. All software shall be delivered via CD/DVD ROM disk(s) and included with the system.

3.3.14 Freeware, shareware, and open source software. Software shall not use any freeware or shareware. Software shall avoid use of open source software. If open source software is used, it shall be supported with a Configuration Management Plan and Support Plan.

3.3.15 High level vulnerabilities. Software shall not have any high-level vulnerabilities as

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:32Z
Check the source to verify that this is the current version before use.

reported in the NIST National Vulnerability Database (https://nvd.nist.gov/vuln) concerning Common Vulnerability and Exposures (CVEs).

3.3.16 Root directory. Software shall not write to the system root directory. Functionality of the application software shall not require administrative privileges.

3.3.17 Adobe Flash. Software shall not use Adobe Flash.

3.3.18 Self-signed certificates. Software shall not use self-signed certificates.

3.3.19 Foreign software. Software shall not use foreign software of any kind.

3.3.20 Image-acquisition software. The software shall provide a means to record pertinent data related to the image including, the ability to record and organize file data per ASTM E2339, Digital Imaging and Communication in Nondestructive Evaluation (DICONDE) and E2738, Digital Imaging and Communication in Nondestructive Evaluation (DICONDE) for Computed Radiography (CR) Test Methods. The proposal shall include a conformance statement that accurately describes DICONDE implementation. Any deviations from DICONDE standards shall be identified in the proposal.

3.3.20.1 Metadata. Metadata shall be automatically recalled with the image during subsequent reviews.

3.3.20.2 Read/erase. Acquisition software activates the start of the read/erase cycle.

3.3.21 Image leveling. The CR system image analysis software shall be capable of leveling the image in no more than two operations.

3.3.22 Software bit depth. All image manipulations and calculations shall be conducted with linear data at least 14-bit grayscale pixel depth.

3.3.23 Image analysis software. Image analysis software shall provide the following tools as a minimum:

3.3.23.1 Window/level. The ability to manually adjust the Window (contrast) and Level (brightness) independently and using a region of interest (ROI) tool.

3.3.23.2 Pan. The ability to manually pan the image.

3.3.23.3 Magnification. The ability to manually adjust the magnification (zoom) up to a minimum of 400 percent, including the ability to display the image at 1:1 pixel mapping.

3.3.23.4 Thumbnail views. The ability to display a thumbnail view indicating the displayed portion of the image relative to the entire image when using digital magnification.

3.3.23.5 Image zoom. The ability to display the digital magnification (zoom) level.

3.3.23.6 Magnifier tool. The ability to magnify an area of interest with a “floating” magnifier tool to 300 percent magnification.

3.3.23.7 Simultaneous viewing. The ability to display at a minimum of four images simultaneously with independent software feature control of each image.

3.3.23.8 Gray scale readout. The ability to measure and display the gray scale value (pixel value) of individual pixels.

3.3.23.9 Measurements. The ability to obtain calibrated linear, area, and angle, measurements with user selectable metric or English format, and the ability to compensate for any geometric enlargement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:32Z
Check the source to verify that this is the current version before use.

3.3.23.10 Region of interest (ROI). The ability to generate a ROI. This tool shall display the size of the ROI (e.g. x pixels by y pixels), and at a minimum display the statistical mean, standard deviation, and signal-to-noise ratio of the pixel values in the region of interest.

3.3.23.11 Histogram. The ability to generate a histogram of pixel values (gray scale value) along a line or ROI tool providing at a minimum, minimum and maximum pixel value, average pixel value, and standard deviation of pixel values.

3.3.23.12 Image manipulation and enhancements. The system shall have the capability to initially display images without image enhancements and then allow the user to apply filters, edge enhancements or contrast enhancements. The system shall provide:

a. The ability to display the image in either negative or positive polarity.

b. The ability to change image display orientation, including the ability to mirror and rotate the image at 90-degree intervals.

c. The ability to include user generated annotations on the image including but not limited to text, circle/box outlines, and arrows.

d. The ability to display any selected DICONDE data on the image.

e. The ability for the user to select image processing filters for inspection.

f. The ability to display the image menu filters applied to the image.

3.3.24 Display-monitor software. The display monitor shall provide the following capabilities:

3.3.24.1 SMPTE RP133 pattern. Monitor software shall include the SMPTE RP133 test pattern, and the ability to display the test pattern on the high-resolution monochrome display monitor.

3.3.24.2 Self-diagnostic capability. Monitor software shall include self-diagnostic software, including but not limited to, display brightness checks on the high-resolution monochrome display monitor.

3.4 Database, data archiving and data sharing. The CR system database shall provide for user selectable display of DICONDE tags and for searching and sorting of images by selectable DICONDE tags.

3.4.1 Filing structure. The CR system shall be able to logically search image inventory for image selection and retrieval.

3.4.2 File format. Image files shall be in a DICONDE compliant file format per ASTM E2339 and ASTM E2738. Any deviations shall be identified in the proposal.

3.4.2.1 DICONDE headers. The following data shall be stored in the DICONDE headers:

a. CR system manufacturer and model

b. Image analysis software and version number

c. Imaging plate serial number or other unique identifier

d. Pixel pitch (in microns)

e. Unit name (nomenclature)

f. Aircraft identification

g. Component identification

h. Technique parameters (kV, mA, time, distance)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:32Z
Check the source to verify that this is the current version before use.

i. Other notes as required

j. Date of inspection

k. Inspecting activity

3.4.2.2 Other file formats. Image files shall be able to be saved and exported at a minimum in lossless image format such as PNG, GIF, and TIFF.

3.4.3 Data saving/archiving. Software shall retain the original (raw) data. The original data shall be capable of being re-processed and re-displayed using the same or another process algorithm without having to take another radiographic exposure.

a. The full bit depth of the image, as read from the IP, shall be retained in the saved image.

b. Digital annotation shall not permanently alter the nature of the image or hinder interpretation of an area within the image.

3.4.4 Data sharing.

a. The CR system shall be capable of saving and retrieving data directly to/from the system's hard drive, and CDs, DVDs, and external hard drives.

b. The CR system shall be capable of importing and exporting images to/from other DICONDE compliant workstations.

c. Images exported to other DICONDE network shall be able to be viewed, manipulated, interpreted, and stored.

d. Users shall be able to transmit DICONDE images through DICONDE network without loss of image resolution when viewed on comparable display screens.

e. The CR image analysis software shall have the ability to allow viewing of third-party DICONDE images, without requiring a system workstation.

3.4.5 External image viewing software. Software for viewing on a non-workstation shall be provided.

a. External image viewing software shall include at least the following tools: window/level, magnification, and pan.

b. No special, proprietary software or workstation shall be required to operate the external image-viewing software.

c. External image-viewing software shall be self-executing either from portable media or by copying to a third-party personal computer.

d. External image-viewing software shall be created or provided at a minimum on a CD/DVD.

3.5 Diagnostics. The software shall enable the user to perform system diagnostics.

3.6 Calibration. No factory recalibration decals shall be applied to the instrument.

3.7 DICONDE image. A DICONDE compliant image of the line pair gauges of the Process Control Phantom described in paragraph 3.8.2 shall be transmitted to the AF NDI Office at time of proposal submission to confirm compliance. File size is limited to 10MB. Alternatively, the proposal can include a website URL to download the image file, or a CD-ROM may be sent to the Cognizant Engineer. Notional data shall be stored under the DICONDE headers.

3.8 Accessories.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:32Z
Check the source to verify that this is the current version before use.

3.8.1 Shipping containers. The CR system shall include hard-shell carrying cases and be suitable for storing and shipping the unit with accessories.

3.8.1.1 Padding. Carrying cases shall be appropriately padded to minimize potential shipping damage to the complete systems.

8.1.2 Drop test. All proposals shall include testing conducted by an independent laboratory IAW the procedure described in MIL-PRF-28800F, paragraph 4.5.5.4.2, or equivalent, for a 46 centimeter (18 inch) drop in the hard cases.

3.8.1.3 Wheels and handles. Cases shall have wheels and handles. (Note: it is desirable for cases be of similar size to facilitate stacking and palletizing for shipment.)

3.8.1.4 Prior drop testing. Prior testing accomplished to a previous revision of MIL- PRF-28800 is acceptable, but shall be identified in the proposal.

3.8.2 Process control devices. The CR system shall include an ASTM E2445/E2445M Type II Process Control Phantom. NSN 6635-01-555-4372 (CAGE 6KCS9, ASES LLC, DBA FIELD AEROSPACE, CP 405-219-3420, http://www.fieldaero.com**,** P/N A0295004002) is the initial configuration and drawings are available upon request. It is available commercially at CAGE 3DF34, NDT SUPPLY, CP 913- 685-0675, http://ndtsupply.com/kowotest-usaf-cr-phantom-panel.html, P/N RT-KT-CR-Phan- USAF. The CR System shall also include ASTM E 746 aluminum Equivalent Penetrameter Sensitivity (EPS) plaques and 0.75 inch thick aluminum absorber plate; the USAF approved supplier is CAGE 29884, RAY CHECK MANUFACTURING, CP 559-292-0115, http://www.ray-check.com/radiographic.html, P/Ns RIQR2024 and RIOI-AP. Other suppliers require approval by the Cognizant Engineer.

3.8.3 Image quality indicators. The CR system shall contain the following image quality indicators: a full ASTM-E1742/E1742M , Annex A1 plaque-type IQI set meeting the requirements of NSN 6635-00-185-1983, P/N ASTM E1742 (Air Force Master Set with AL, CU, FE, MG, SS, TI) and ASTM E-747 wire IQI sets A-B, in aluminum, magnesium, and stainless steel, and A-B in Inconel and titanium.

3.8.4 Documentation. One paper copy and one electronic copy of the commercial operation and maintenance instructions, including calibrations procedures and adjustments, as applicable, shall be shipped with each system. An electronic copy shall be loaded on the workstation. All documentation shall be releasable for third-party repairs. A draft copy shall be provided with the production evaluation article for engineering approval. Two paper and two electronic copies of the final version shall be sent to the USAF Equipment Specialist and Cognizant Engineer. The CR system's Operation and Technical Manual shall at a minimum include:

a. CR System Technical Operation and Maintenance.

b. CR System Illustrated Parts Breakdown including compatible IPs available commercially.

c. Service and Calibration Maintenance Schedule Requirements.

d. Listing of all available accessory items and services, warranty information, and vendor/contractor's point of contact.

e. Safety Data Sheets for items as applicable.

f. See data item descriptions DI-TMSS-80527A and DI-MISC-80557 for additional information.

3.9 Process control images. Radiographic images of the serialized process control devices described in paragraph 3.8.2, which are actually delivered with each system, shall be included on

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:32Z
Check the source to verify that this is the current version before use.

the workstation to document the manufacture's quality control processes. The image shall contain serial numbers of the process control devices, the workstation, scanner, and date formed using metallic lead letters and numbers, or hand-scribed on metallic lead foil.

TABLE I. CR System Component Summary.

Component	Description	Quantity	Paragraph
Imaging plate	General Purpose 14 x 17 inch	2	3.3.4.2
Imaging plate	General Purpose 8 x 10 inch	2	3.3.4.2
Imaging plate	General Purpose 5 x 7 inch	2	3.3.4.2
Imaging plate	High resolution 14 x 17 inch	5	3.3.4.3
Imaging plate	High resolution 8 x 10 inch	2	3.3.4.3
Imaging plate	High resolution 5 x 7 inch	2	3.3.4.3
Imaging plate	Ultra-high resolution 8 x 10 inch	2	3.3.4.4
Imaging plate	Ultra-high Resolution 5 x 7 inch	2	3.3.4.4
Flexible cassette	14 x 17 inch	7	3.3.4.5
Flexible cassette	8 x 10 inch	6	3.3.4.5
Flexible cassette	5 x 7 inch	6	3.3.4.5
IP cleaning system	Fluid, wiping cloths, etc., for 100 IPs	1	3.3.5
Maintenance kit	Maintenance kit	1	3.3.6
Scanner	Image reader/erasure unit	1	3.3.7
Cut IP adapter	Scan cut IPs	2	3.3.7.f
Processing cassette	Processing cassette	2	3.3.7.g
Workstation	Scanner control and file imaging	1	3.3.8
Display monitor	Workstation monitor	1	3.3.8, 3.3.9
Software	Software for system operation and file imaging	1	3.3.10
Process control images	Process control device image	1	3.9
Shipping containers	System shipping containers	1 set	3.8.1
Process control device	Process Control Phantom. NSN 6635-01-555-4372, EPS plaques, absorber plate	1 ea	3.8.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:32Z
Check the source to verify that this is the current version before use.

TABLE I. CR System Component Summary.

Component	Description	Quantity	Paragraph
Image Quality Indicators	ASTM-E1742/E1742M plaque-type penetrameter set, NSN 6635-00-185-1983 and ASTM E-747 wire IQI sets	2 sets	3.8.3
ASTM E2445/E2445M report	Report and images	1	3.10
Manual	Operation and maintenance manuals, paper and electronic	1 ea system, 2 ea for approval	3.8.4
Software	CD/DVD ROM	1	3.3.13
Market acceptability	Report/statement	1	5.10

3.10 Test report. A report based on ASTM E2445/E2445M -14, or later versions with approval, evaluations, along with images used for evaluation shall be provided using a production system and IPs. The following apply to this report.

a. The X-ray generator shall have a maximum focal spot of 1.5 mm (EN12543).

b. A Type II CR phantom shall be used where applicable.

c. Tests shall be performed for low-energy applications.

d. Tests shall be performed using one (1) IP from two (2) different production lots or batches for each IP type.

3.10.1 Core image quality tests. Tests defined in ASTM E2445/E2445M, Test Procedures, Core Image Quality Tests, 8.1.1.1 through 8.1.1.12 shall be performed using high resolution IPs supplied per paragraph 3.3.4.3 and the ASTM E2445/E2445M Type II test phantom unless otherwise noted. Tests 8.1.1.1 through 8.1.1.11 shall be performed per Procedure for Core Image Quality Tests, 8.2. Tests in 8.1.1.12 for SNR shall be performed per Procedure for Sensitivity Tests Considering Image Noise, 8.4.

3.10.2 Erasure test. Testing defined in ASTM E2445/E2445M, Erasure, 8.3.1 shall be performed five (5) times using a 14 x 17 inch high resolution IP supplied per 3.3.4.3.

3.10.3 Phantom test. Results per ASTM E2445/E2445M, sections 9.3.1 and 9.3.3 through 9.3.9.

4 REGULATORY REQUIREMENTS.

4.1 Recoverable materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practical, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.1.1 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pounds test sets, provided that they fall within specified tolerances using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this CID are met. If a product is manufactured to metric dimensions and

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:32Z
Check the source to verify that this is the current version before use.

those dimensions exceed tolerances specified in the inch-pound test sets, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

4.1.2 Toxic chemicals, hazardous substances, and ozone-depleting chemicals. The use of toxic chemicals, hazardous substances, or ozone-depleting chemicals shall be avoided, if feasible.

5. PRODUCT CONFORMANCE. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.1 Production evaluation. The CR system (i.e. scanner, workstation, IPs, display monitor) shall pass any of the production evaluations specified as follows using an Air Force approved X-ray apparatus. Final image evaluation shall be performed without digital image processing (software filters).

5.2 Spatial resolution and SNR. The achieving of spatial resolution, and percent EPS for delivered IPs noted in Table II.

TABLE II. Spatial Resolution (Visual) Requirement Based on EPS Performance.

IP Type	Line profile tool (visual)	EPS percent
General purpose	2.5	N/A
High resolution	6.0	1.28% or less
Ultra-high resolution	8.0	1.28% or less

5.3 B-52 imaging. Successful completion of T.O. 1B-52H-36, wing trailing edge flaptrack inspection by scanning cut IPs and visualizing the area of interest according to the T.O. and the 2-2T hole (or wire equivalent) in the image quality indicator in the area of interest.

5.4 F-16 imaging. Successful completion of T.O. 1F-16A-36, wing internal spar, wing tip rib, and front spar by visualizing the area of interest according to the T.O. and the 2-2T hole (or wire equivalent) in the image quality indicator in the area of interest.

5.5 T-38 imaging. Successful completion of T.O. 1T-38C-36 wing spar and ribs by visualizing the area of interest according to the T.O. and the 2-2T hole (or wire equivalent) in the image quality indicator in the area of interest.

5.6 F-15 imaging. Successful completion of T.O. 1F-15A-36, tip closure rib (P/N 68A115105, without straps) for cracks, L/R first ramp rib support (P/N 68A327122), and L/R horizontal stabilator inboard leading edge rib (P/N 68A210130) for cracks by visualizing the area of interest according to the TO and the 2-2T hole (or wire equivalent) in the image quality indicator in the area of interest.

5.7 Weld certification tube. Successful radiographic inspection of a government owned welder certification tube butt joint weld specimen that contains porosity weld defects using existing T.O. 00-25-252 procedures by visualizing defects according to the T.O. and the 2-2T hole in the image quality indicator in the area of interest.

5.8 Fatigue specimens. Successful radiographic inspection of a predefined set of government owned fatigue crack specimens, imaging 5 of 15 cracks using high resolution IPs.

5.9 Market acceptability. All of the following market acceptance criteria are necessary to

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:32Z
Check the source to verify that this is the current version before use.

document the quality, reliability, and supportability of the product to be provided under this specification.

a. The company producing the item must have been producing CR systems for at least the past three years.

b. The company must have sold similar commercial systems in the industrial market for a minimum of two years. Upgraded systems less than two years old similar to previous systems may be proposed.

c. The company must have sold at least 150 instruments similar to the instruments being offered, with no major defects within the last three years. A major defect is a defect, which affects the performance of the instrument and requires an engineering change to the instrument's original configuration in order for the instrument to perform the requirements of this document.

d. Instrument repairs shall be provided in the contiguous U.S. and shall have an average repair time of two calendar weeks or less.

e. Data shall be provided showing the system is capable of operation for at least 1500 hours in a one year period with no major repairs for 50 systems.

6 PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7 NOTES.

7.1 Sources of documents.

a. Copies of military specifications, standards, data item descriptions, and handbooks referenced herein are available at https://quicksearch.dla.mil.

b. Copies of American Society for Testing and Materials (ASTM) documents are available at https://www.astm.org/.

c. Copies of Society of Motion Picture and Television Engineers (SMPTE) documents are available at https://www.smpte.org/.

d. Copies of Technical orders (TO) are available at https://etims.cce.af.mil/ETIMS/index.

e. Copies of the Federal Acquisition Regulation (FAR) are available at https://www.acquisition.gov/browse/index/far.

7.2 National Stock Numbers (NSNs). The generic NSN for this equipment is 6635-01-523-5766.

7.3 Ordering data. The contract or order should specify the following:

a. Title, number, and date of this document.

b. Product conformance provisions.

c. The solicitation will provide for any required independent-laboratory pre-production testing, or Government testing, as appropriate.

7.4 Key words.

Imaging Plate
Storage Phosphor
DICONDE (Digital Imaging and Communication Nondestructive Evaluation)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:32Z
Check the source to verify that this is the current version before use.

MILITARY INTEREST

Custodian
Army – GL
Navy – AS
Air Force – 171

Review Activities
DLA – GS
Air Force – 20

Preparing activity
Air Force – 171

Agent:
AIR Force – 110

(Project 6635-2021-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Because organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:32Z
Check the source to verify that this is the current version before use.